exclurad_py: exact radiative corrections and target straggling for exclusive meson electroproduction

Design, validation and the first π^0 production

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Abstract

`exclurad_py` is an event generator for exclusive meson electroproduction $ep \to e' h X$ covering π^0, η, ρ, ϕ and J/ψ in one code, with exact first-order QED radiative corrections built on EXCLURAD and with pre-vertex external radiation in the target. This note is the evidence. The central result is that the generated event density reproduces the exact calculation bin by bin without reweighting: over the azimuth, where the correction swings from 1.15 to 0.92, events and quadrature agree with $\chi^2/N = 0.79$ over eighteen bins. The target radiator is validated against an independent implementation, exclugen, to $0.6\,\sigma$. Four independent samples on three machines, two processor families and two python generations are statistically indistinguishable. The note states, with equal care, what is *not* yet established: the vector channels have no external anchor for their radiative correction, and ρ and ϕ still lack the Breit–Wigner mass, which for ρ is not a cosmetic omission.

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1 Scope and status

The five channels are at different stages, and the difference matters more than any single number in this note.

channel	internal RC	external anchor	B–W mass	status
π^0	validated	published EXCLURAD table	n/a	production running
η	validated	published table via v_{cut}	n/a	next
ρ	vs. own quadrature	none	**missing**	blocked on B–W
ϕ	vs. own quadrature	none	**missing**	blocked on B–W
J/ψ	vs. own quadrature	none	not needed	box needs care

"Validated against our own quadrature" is a weaker statement than it sounds: generator and reference share the same `rc_factor`, so an error inside that function is invisible to the comparison. It catches sampling errors, which are the common ones, and cannot catch a wrong core. Only π^0 and η have something outside the code to lean on. This is why they come first.

2 The radiative correction

EXCLURAD computes, in the Bardin–Shumeiko formalism, the observed cross section for $ep \to e'h + X_{\mathrm{miss}}$ where the undetected system is reconstructed by missing mass. The variable is the inelasticity

$$v \;=\; M_X^2(ep \to e'h) - m_U^2, \tag{1}$$

zero for a non-radiative event. With an exclusivity cut $v < v_{\mathrm{cut}}$,

$$\sigma_{\mathrm{obs}}(v_{\mathrm{cut}}) \;=\; \sigma_{\mathrm{Born}}\,\big(1 + \delta(v_{\mathrm{cut}})\big), \tag{2}$$

and δ depends on all four hadronic variables and on the cut. Inside the code the same quantity is stored as the multiplicative factor $\eta = \sigma_{\mathrm{obs}}/\sigma_{\mathrm{Born}}$; the note quotes δ in that convention, so $\delta \approx 0.9$ means a 10 % suppression, not a 90 % one.

The soft and hard parts are separated at an arbitrary v_{min}. Nothing physical may depend on it, and §6 shows that nothing does.

3 How the correction is extracted from events

Two runs of the same card differing only in `iborn` give

$$\delta(\mathrm{bin}) = \frac{N_{\mathrm{rad}}(\mathrm{bin})/N_{\mathrm{rad}}}{N_{\mathrm{born}}(\mathrm{bin})/N_{\mathrm{born}}} \times \frac{\sigma_{\mathrm{obs}}}{\sigma_{\mathrm{Born}}}. \tag{3}$$

This is the whole method. Both samples come from the same generator, so the ratio isolates the radiative machinery. Born events cost about a tenth of a radiative one, so the Born sample is made five times larger and contributes about a fifth of the variance.

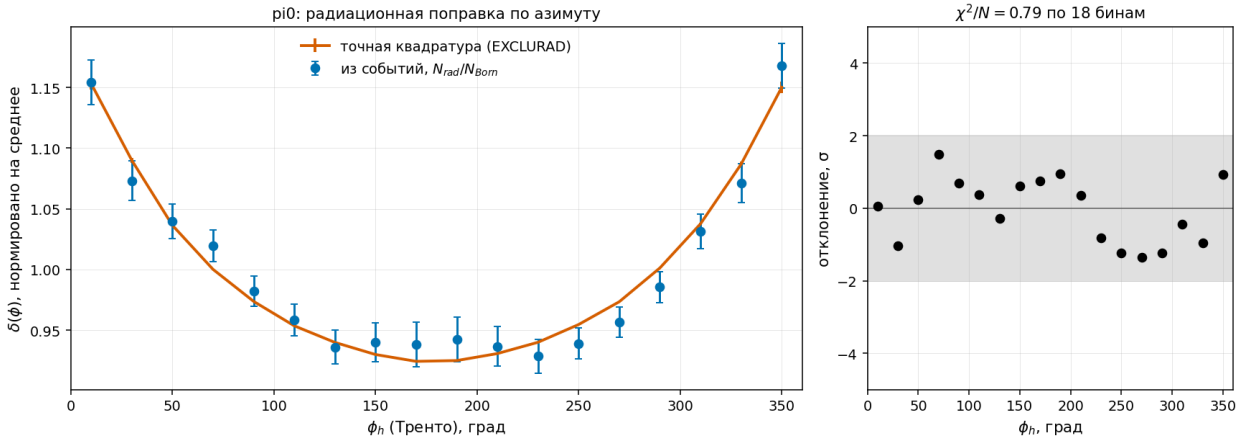


Figure 1: π^0, $E = 10.6$ GeV, Q^2 1.5–2.5, W^2 5.25–6.75, $|t|$ 0.17–2.0, $v_{\text{cut}} = 0.20$ GeV^2, no target radiator. Left: the radiative correction over the azimuth obtained two independent ways — from the events as the ratio of two histograms (points), and from direct integration of the Bardin–Shumeiko formulas at the same kinematic points (line). Right: the difference in standard deviations. 100k radiative and 500k Born events; the quadrature uses 20k points and carries its own error, which is included in the pulls.

4 Proof: the correction over the azimuth

The azimuth is the sharpest test available. The correction varies more over ϕ than over any other variable, and ϕ is where a convention error — Trento or not, which hadron defines the plane, the sign of σ_{LT} — shows up. Integrated quantities hide exactly these mistakes.

Figure 1 is the central result. The two determinations share no machinery: one counts events, the other integrates formulas. They agree with $\chi^2/N = 0.79$ over eighteen bins, the largest deviation being $1.5\,\sigma$, across a correction that swings by a quarter.

The practical consequence is that an analysis binning our events in ϕ recovers the exact correction by construction — no reweighting, no correction table.

ϕ (deg)	from events	from quadrature	σ
10	1.1542 ± 0.0185	1.1532 ± 0.0046	0.1
90	0.9822 ± 0.0125	0.9737 ± 0.0003	0.7
170	0.9379 ± 0.0184	0.9242 ± 0.0005	0.7
250	0.9390 ± 0.0127	0.9546 ± 0.0003	-1.2
350	1.1678 ± 0.0185	1.1501 ± 0.0048	0.9

5 The same in the other variables

Over the full production box the agreement with quadrature holds on every axis. Measured on the 100k union, maximum deviation in standard deviations over twelve bins per axis:

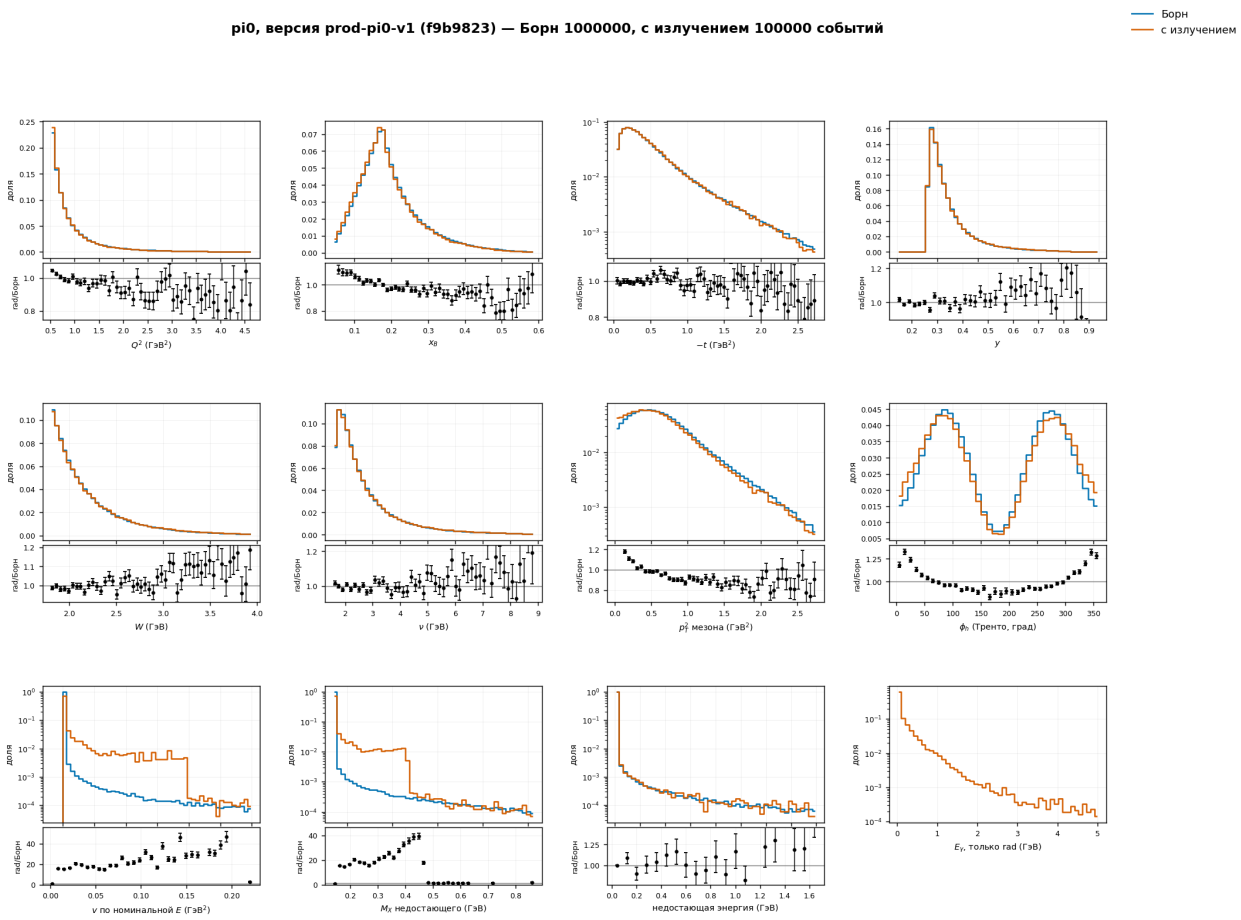


Figure 2: π^0 production sample, radiator on. Upper half of each cell: Born and radiative distributions normalised to unit area; lower half: their ratio. The ratio is the correction as a function of that variable. Note the inelasticity and missing-mass panels: the Born tail stops at the cut while the radiative one extends, which is the radiative tail itself.

channel	Q^2	W	$\lvert t\rvert$	ϕ
π^0	1.8	1.9	1.7	2.0
η	2.0	1.6	1.9	2.6
ρ	2.7	1.5	2.8	1.6
ϕ	2.1	2.7	1.4	1.9
J/ψ	2.9	3.0	1.7	1.9

At twelve bins the worst of twelve is expected around $2.5\,\sigma$ purely by chance, so these numbers are what agreement looks like.

6 Independence of the soft/hard separator

$v_{\min}$ splits the tail integral into an analytic soft part and a numerical hard part. Moving it by two orders of magnitude moves half the contribution from one to the other, and the total must not notice.

$v_{\min}$ (GeV^2)	σ_{obs} (nb)	hard fraction
10^{-5}	0.026400 ± 0.000088	51.3 %
10^{-4}	0.026121 ± 0.000068	42.1 %
10^{-3}	0.026236 ± 0.000087	32.5 %

The hard fraction changes by a factor of 1.6; the cross section is constant to 1.1 %, with the largest pairwise difference at $2.5\,\sigma$. There is no monotonic trend — the middle value lies below both ends — so this looks like statistics rather than a residual dependence. It is honest to say that $2.5\,\sigma$ is "not caught" rather than "clean"; separating a half-percent systematic from a fluctuation needs four times the statistics.

7 External radiation in the target

An electron crosses part of the target before interacting and radiates on the way in. Geant4 cannot supply this: it starts at the vertex and takes the beam energy as given, so the incoming leg is the generator's responsibility and nothing in the CLAS12 chain was providing it. The outgoing leg is deliberately *not* modelled here — Geant4 does it, and doing it twice is worse than not doing it at all.

The sampling follows exclugen and aao_rad: the vertex is uniform over the cell, $bt = (4/3)(v_z - v_{z,\min})/X_0$ with $X_0(\mathrm{LH_2}) = 890.4$ cm, and the surviving energy fraction is $u^{1/bt}$. Against the analytic straggling law over 400k draws: $P(\text{loss} > 1\,\%)$ is 1.674 % against 1.704 %, and $P(\text{loss} > 10\,\%)$ is 0.818 % against 0.857 %.

Validation against an independent implementation. The observable effect is that events are pushed out of the exclusivity window, because an analysis reconstructs the missing mass with the *nominal* beam energy. Running exclugen twice, with and without its own radiator, over six independent pairs of 600k events gives a shift of 0.873 ± 0.112 % in the cut efficiency. Our generator, measured the same way:

source	fraction above the cut	events
mac mini	$0.865 \pm 0.047\,\%$	40k
JLab farm	$0.771 \pm 0.031\,\%$	80k
combined	$\mathbf{0.800 \pm 0.026}\,\%$	
exclugen (reference)	$0.873 \pm 0.112\,\%$	6×600k

Agreement at $0.6\,\sigma$, and with the radiator off, exactly zero events out of forty thousand lie above the cut. This is the first piece of our physics checked against somebody else's code rather than against our own quadrature.

One measurement lesson is worth recording. The scatter between exclugen pairs (0.275 %) exceeds the binomial error of a single pair (0.177 %), because exclugen rebuilds its sampling ceiling from a prescan each run. The first two pairs gave -1.20 and $-0.44\,\%$ and disagreed by three standard deviations; both were correct and both were useless alone.

What this means for the user of the sample. With the radiator on, v_{cut} is applied in *true* variables. About 0.8 % of the delivered events will fail an exclusivity cut made with the nominal beam energy — for Born events too, since the electron loses energy before the vertex whatever happens there. The sample is nevertheless complete:

$$v_{\mathrm{rec}} - v_{\mathrm{true}} = 2\,\Delta E\,(E_X - p_{X,z}) \;\geq\; 0, \tag{4}$$

so external loss can only move events out of the window, never into it.

8 Reproducibility

The worst χ^2/N over ten variables is 1.45 (p_T^2) and the median is about 1.1. Nothing in the chain — processor, library version, interactive versus batch — leaves a trace in the sample. Two further checks: a run repeated with the same seed reproduces its output byte for byte, and 100k events pooled from two machines contain exactly 100k distinct events.

9 Output integrity

Conservation laws are checked per event rather than inferred from the shape of a distribution: the header's particle count, four-momentum against beam plus target over the status-1 particles, mass shells, the written mass against PDG, decay closure, charge, and the beam-polarisation column. All five channels pass; four-momentum closes to 1.25×10^{-8} GeV and the decay to 10^{-10}, both far inside the 10^{-6} tolerance.

10 What is not established

The vector channels have no external anchor. ρ, ϕ and J/ψ are checked against our own quadrature only. DIFFRAD, the Akushevich code for vector mesons, is on disk and is the natural reference; this comparison has not been made.

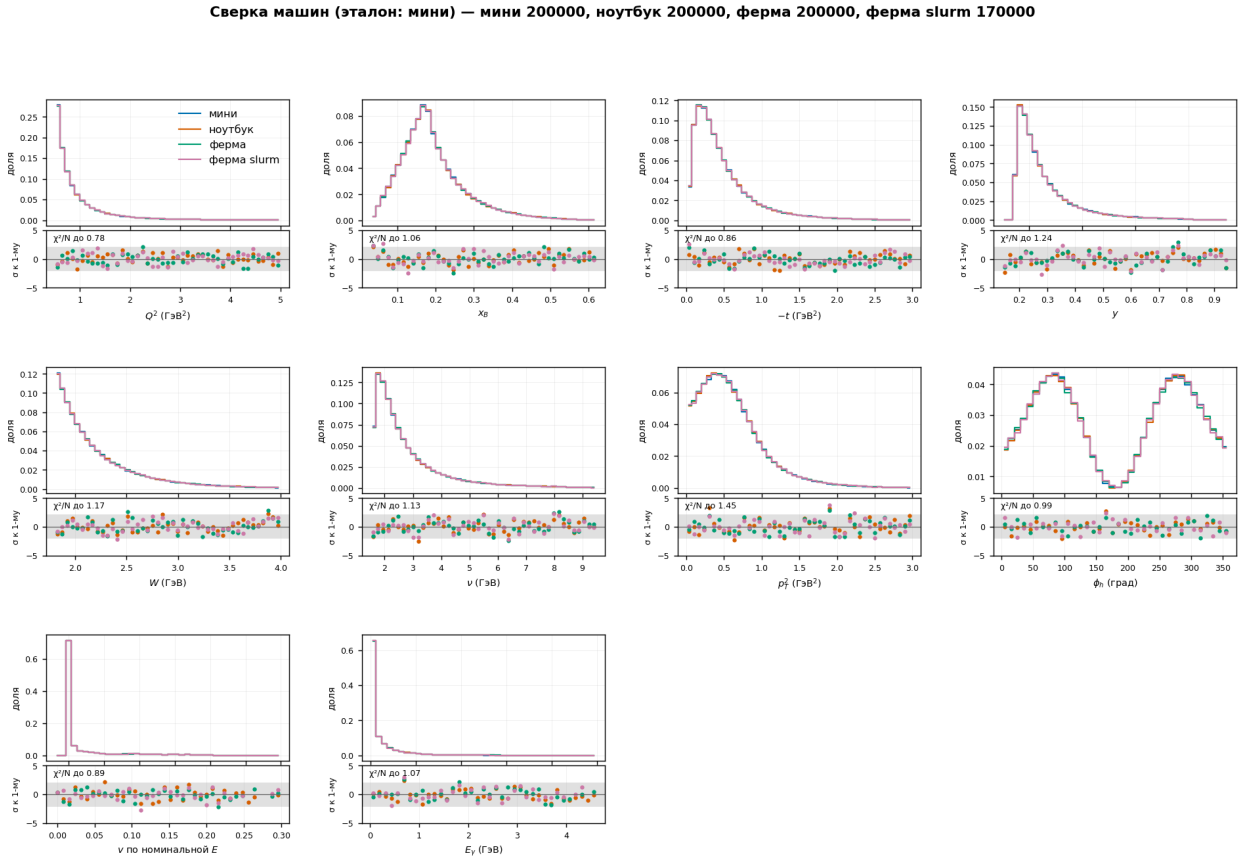


Figure 3: Four independent samples: mac mini (M4), MacBook Pro (M3 Max), the JLab farm interactively and the same farm through slurm. Different seeds, different processors, python 3.9 with numpy 1.23 against python 3.14 with numpy 2.4. Lower panels: deviation from the first sample in standard deviations, grey band $\pm 2\sigma$.

ρ and ϕ have no Breit–Wigner mass. The meson is generated exactly on the pole. For ϕ, $\Gamma/M = 0.4\,\%$ and this is below resolution; for ρ, $\Gamma = 150$ MeV at $M = 775$ MeV, and a fixed mass is wrong in a way that matters. The mass enters the t limits, the W threshold and through them the region where the correction is evaluated, so this is not only the shape of the peak. The sampler has been ported from `diffrad_vm.f90` and agrees with it ($\chi^2/N = 0.80$ for ρ, 0.73 for ϕ over 200k draws each); what remains is to make the mass a property of the event throughout the kinematics.

ρ and ϕ run with no inelasticity cut. $v_{\mathrm{cut}} = 0$ by decision, meaning the tail runs to the kinematic limit where the diffractive model is extrapolated far outside its fit. With $v_{\mathrm{cut}} = 0$ the ρ correction carries an $8\,\%$ slope in Q^2 that the quadrature does not have ($\chi^2/N = 4.04$); at $v_{\mathrm{cut}} = 0.20$ the slope is gone (2.04).

π^0 table interpolation. The interpolation rms is 0.029 against a tolerance of 0.02, and refining does not help: $1000 \to 6144$ nodes moves it from 0.029 to 0.027, while the other four channels sit at 0.002–0.006. Correctness is unaffected — the second stage is exact — but efficiency is.

The exponentiation systematic is measured but not resolved: δ with and without the YFS exponentiation differs by $-0.66\,\%$ (π^0), $-0.48\,\%$ (η), $-1.47\,\%$ (ρ), $-1.44\,\%$ (ϕ) and $-0.09\,\%$ (J/ψ). It should be quoted as a theory systematic.

11 Summary

For π^0 the generator is ready: the event density reproduces the exact calculation over the azimuth to $\chi^2/N = 0.79$, the target radiator agrees with an independent implementation to $0.6\,\sigma$, four samples on three machines are indistinguishable, and the output satisfies the conservation laws event by event. η shares the machinery and needs its own production and its own published-table comparison. The vector channels need the Breit–Wigner mass first — mandatory for ρ and ϕ — and then an external anchor, for which DIFFRAD is available and unused.